FACULTY	Agriculture, Engineering and Natural Sciences		
SCHOOL	Engineering and the Built Environment		
DEPARTMENT	Mechanical and Metallurgical Engineering		
SUBJECT	ENGINEERING MECHANICS II		
SUBJECT CODE	I3641NM		
DATE	26th June, 2024		
DURATION	2 Hours	MARKS	100

First Opportunity Examination

Examiner: Prof. I. O. Ohijeagbon

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External Moderator: Prof. M Waheed (FUNAAB-NIGERIA)

This question paper consists of **3 pages** including this front page.

Instructions

1. Closed book examination
2. Read the questions carefully
3. This question paper consists of **Five** questions
4. Answer **any 4 Questions**
5. Each question carries **25 marks**
6. Neatly drawn sketches should be used where required

Question One

(a) An investigation was carried out to study the rectilinear kinematic motion of a particle. Illustrate the possible analysis you would likely consider to describe the characteristics of the motion if the acceleration is changing with time. **[8 Marks]**

(b) A small projectile is fired vertically downward into a fluid medium with an initial velocity of 60 m/s. Due to the drag resistance of the fluid the projectile experiences a deceleration of $a = (-0.4v^3)\, m/s^2$, where v is in m/s.

(I) Express the (i) velocity and (ii) position of the projectile as a function of time.

(II) Determine the projectile's (i) velocity and (ii) position 5 s after it is fired.

(III) What would be the expected projectile's (i) velocity and (ii) position 5 s after it is fired, if it experiences an acceleration $a = 2.5g$, where $g =$ acceleration due to gravity.

[17 Marks]

Question Two

(a) The free-flight motion of a projectile is illustrated as shown in Figure Q. 2 (a), launched at point (x_0, y_0), with an initial velocity of $\boldsymbol{v}_0$, having components $(\boldsymbol{v}_0)_x$ and $(\boldsymbol{v}_0)_y$. Starting from first principles, conduct the kinematic analysis of a projectile for:

(i) Horizontal motion

(ii) Vertical motion

[7 Marks]

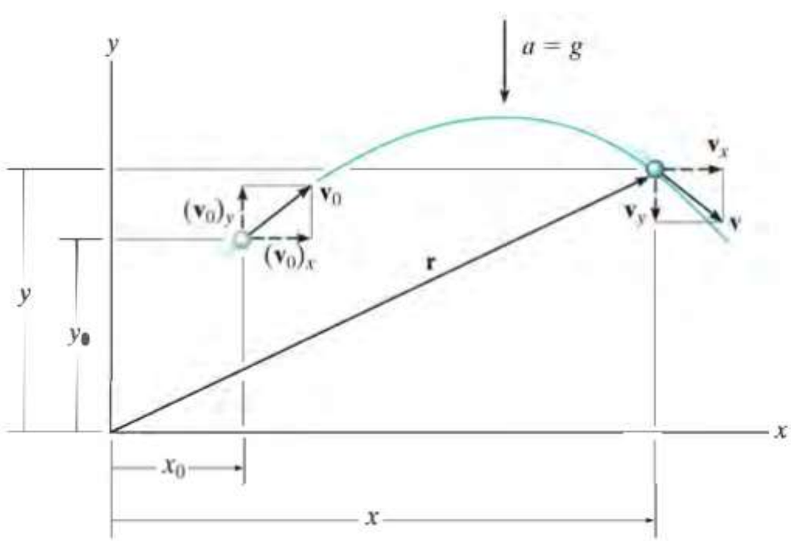

Figure Q. 2 (a)

(b) Pumping of water from a flooded area to a nearby canal over a landscape was required. Water is pumped through the hose into the air at a slope of 30°, from a height of 1 m from the ground. Projected water is suspended in the air before touching the ground after 1.5s. Determine:

(i) the speed at which water exit the water hose,

(ii) the horizontal distance water travels before striking the ground,

(iii) the maximum height attained by projected water, and

(iv) the components of the velocity when the water strikes the ground.

[18 Marks]

Question Three

(a) The pulley of an industrial electric motor has an angular acceleration of $\alpha = (1.5\theta)\ rad/s^2$, where θ is in radians. The pulley has a radius of $0.2\ m$ and assumed to start from rest. Determine:
(i) the magnitude of the velocity and
(ii) the acceleration of a point $\boldsymbol{P}$ located on the edge of the pulley after it has rotated 2 revolutions.

[15 Marks]

(b) If the pulley in Q. 3 (a) drives a wheel of $0.5\ m$ diameter. Determine:
(i) the speed of the wheel $\boldsymbol{Q}$ in revolution per minute (rpm) and
(ii) the acceleration at a point $\boldsymbol{m}$ located at the outermost part of the wheel $\boldsymbol{Q}$.

[10 Marks]

Question Four

(a) The motor winds in the cable with a constant acceleration, such that the 20-kg crate shown in Figure Q. 4 (a) moves a distance $s = 6\ m$ in 3 s, starting from rest. The coefficient of kinetic friction between the crate and the plane is $\mu_k = 0.3$. Determine:
(i) the tension developed in the cable,
(ii) the tension developed in the cable if the plane was completely horizontal, and
(iii) the tension developed in the cable if the plane was completely vertical.

[15 Marks]

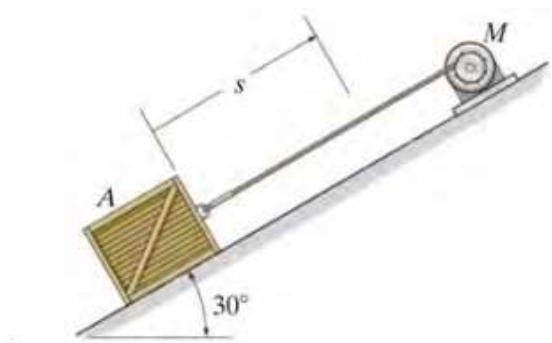

Figure Q. 4 (a)

(b) For the 20-kg crate shown in Figure Q. 4 (a), determine:
(i) the acceleration and velocity of the crate at the inclined angle of 30^0, and
(ii) the acceleration and velocity of the crate if the plane was completely horizontal.

[10 Marks]

Question Five

(a) A car *A* has a mass of 2,000 kg and a center of mass at *G*. The distance of the front and rear wheels from *G* are 1.25 *m* and 0.75 *m* respectively while the height of *G* is 0.3 *m*. If the mass of the wheels are neglected, and the coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.25$. Determine:
(i) the acceleration and
(ii) the reactions at the front and rear wheels, if the rear "driving" wheels are always slipping, whereas the front wheels are free to rotate.

[13 Marks]

(b) Assuming a different car *B* was designed to function conversely to the car *A* in Question 5 (a). If the front "driving" wheels of car *B* are always slipping, whereas the rear wheels are free to rotate. Determine:
(i) the acceleration, and
(ii) the reactions at the front and rear wheels, assuming all other specifications of car *B* are similar to those of car *A*. **[12 Marks]**